

1. TEST CONFIGURATION

Scenario

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Target

TGT-01

Camera

640 x 480

FOV

4 deg x 3 deg

Update Rate

30.0 Hz

Noise: None 10.00 | Camera jitter: OFF 5.00 px/frame | Platform motion: OFF 10.00 px | Turbulence: OFF 10.00 px

2. PERFORMANCE

Processing FPS

118.5

>= 20 FPS

Average Tracking Error

- px

Maximum Tracking Error

- px

<= 10 px

Acquisition Time

- S

<= 2 s

Re-acquisition

- S

<= 1 s

Lock Retention

0.00 %

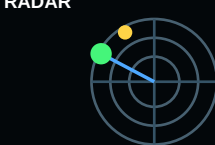
RMSE - px | Target Loss 0.00 % | Avg Processing - ms | Re-acquisitions 0

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Update Rate	30.0 Hz	>= 20 Hz	PASS
Processing Speed	118.5 FPS	>= 20 FPS	PASS
Acquisition Time	- s	<= 2 s	-
Tracking Error (Max)	- px	<= 10 px	-
Target Loss	0.00 %	< 5 %	PASS
Re-acquisition	- s	<= 1 s	-

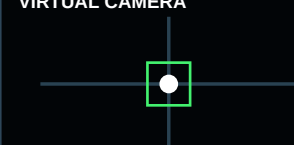
4. VISUAL SNAPSHOT (T = 1.03 s)

RADAR



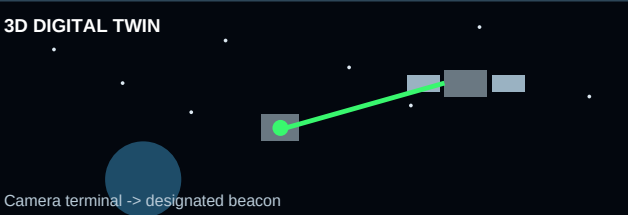
Target / Boresight / Prediction

VIRTUAL CAMERA



Target TGT-01 • Error 342.98 px

3D DIGITAL TWIN



Camera terminal -> designated beacon

State: ACQUIRING | Pan: 1.17 deg | Tilt: 0.78 deg | AI: 0.57

5. SYSTEM SUMMARY

COARSE ALIGNMENT CONTINUES - HANDOFF NOT READY

Designated target TGT-01 at 1.03 s. Detection, AI verification, Kalman prediction and pan/tilt control are reported from the recorded telemetry snapshot.